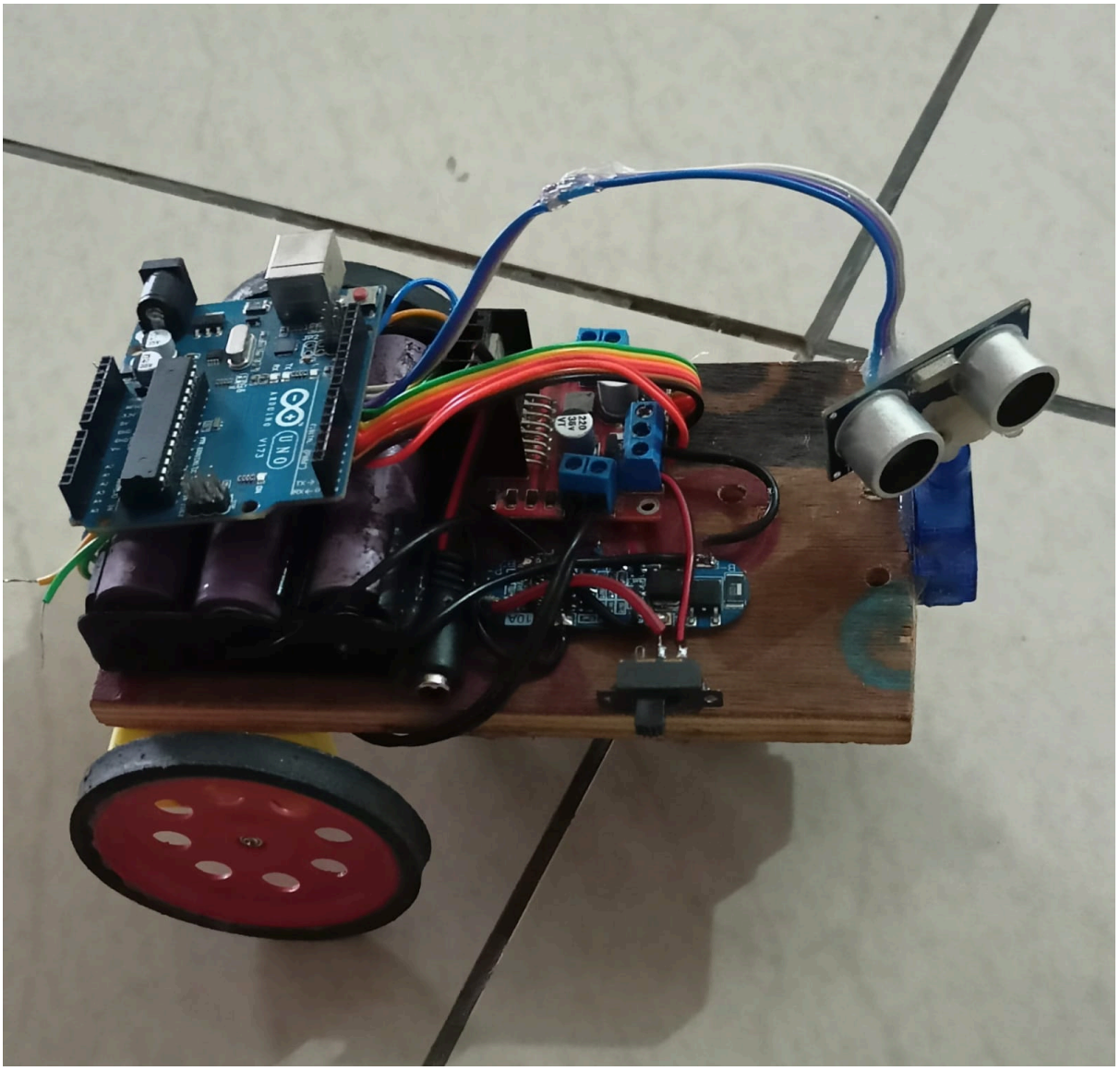
Obstacle-avoiding car

The objective :-

This project is to design and build an autonomous obstacle-avoiding robot car using an HC-SR04 ultrasonic sensor mounted on an SG90 servo motor for dynamic environment scanning. The system detects physical barriers in real time and automatically re-routes the vehicle along a clear path.. The hardware integration ensures reliable power distribution and real-time motor control. Ultimately, this setup demonstrates a practical application of embedded C programming, sensor fusion, and automated path planning for mobile robotics.



Itemized component list :-

- **Arduino UNO R3** (ATmega328P) — Main microcontroller board
- **HC-SR04 Ultrasonic Sensor** — Distance measurement and obstacle detection
- **SG90 Servo Motor** — Pans the ultrasonic sensor left and right to scan pathways
- **L298N Dual H-Bridge Motor Driver Module** — Controls the DC motors and provides regulated 5V power
- **2x DC Gear Motors with Wheels** — Provides robot chassis drive and direction
- **3S Li-ion Battery Pack** — Main power source
- **3S BMS (Battery Management System) Board** — Protects and balances the Li-ion battery pack
- **Robot Chassis Frame & Caster Wheel** — Physical body and passive support wheel
- **Breadboard & Jumper Wires** (Male-to-Male, Male-to-Female) — Wiring and common ground/power distribution
- **100µF – 470µF Electrolytic Capacitor** *(Recommended)* — Placed across the 5V and GND rail to smooth power supply spikes for the servo

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The working principle of this autonomous obstacle-avoiding robot car relies on real-time distance measurement, automated spatial scanning, and dynamic directional decision-making.

1. **Obstacle Detection**: The **HC-SR04 ultrasonic sensor** continuously emits a 40 kHz ultrasonic burst via its transmitter pin. When the wave hits an obstacle, it bounces back and is received by the echo pin. The Arduino Nano measures the pulse return duration t to calculate the distance d in centimeters using the speed of sound:

$$d = \frac{t \times 0.0343 cm/\mu s}{2}$$

2. **Threshold Monitoring**: As the robot moves forward via the **L298N motor driver**, the microcontroller constantly evaluates the distance in front of it. If the path drops below the pre-set threshold ($25cm$), the Arduino immediately signals the L298N driver to stop the motors and briefly reverse the chassis to clear immediate clearance space.
3. **Spatial Scanning**: Once stopped, the Arduino signals the **SG90 servo motor** to pan the ultrasonic sensor 20° to the right and 160° to the left. The sensor captures distance readings for both sides to determine which direction offers the clearest pathway.
4. **Directional Rerouting & Resume**: The microcontroller compares the two scanned distances (d_{right} vs. d_{left}). It commands the L298N driver to pivot the motors toward the side with greater open space, centers the servo back to 90°, and resumes forward movement.

How Pivot Turning Works (Differential Drive)

Your robot car uses two independent DC motors controlled by the L298N dual H-bridge driver. To pivot on the spot, the wheels on opposite sides are driven in **opposite directions at the same speed**:

- **Pivot Right:**
  - **Left Motors:** Drive **FORWARD** (IN1 HIGH, IN2 LOW)
  - **Right Motors:** Drive **BACKWARD** (IN3 LOW, IN4 HIGH)
- **Pivot Left:**
  - **Left Motors:** Drive **BACKWARD** (IN1 LOW, IN2 HIGH)
  - **Right Motors:** Drive **FORWARD** (IN3 HIGH, IN4 LOW)

Why Pivot Turning is Used for Obstacle Avoidance

1. **Zero Turning Radius:** The robot can rotate 180° within its own footprint without needing forward or backward clearance space.
2. **Precision Alignment:** Once the SG90 servo scans and identifies the side with the clearest path (e.g., Left vs. Right), a pivot turn quickly aligns the front distance sensor directly with the new open direction before forward motion resumes.
3. **Simplicity:** It eliminates the need for a complex mechanical steering linkage (like a car's Ackerman steering), relying entirely on motor signal manipulation via code.

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1. Arduino UNO R3 Pinout Connections

Component	Component Pin	Arduino Nano Pin	Notes
L298N Motor Driver	IN1	**D4**	Left Motor Direction 1
	IN2	**D6**	Left Motor Direction 2
	IN3	**D7**	Right Motor Direction 1

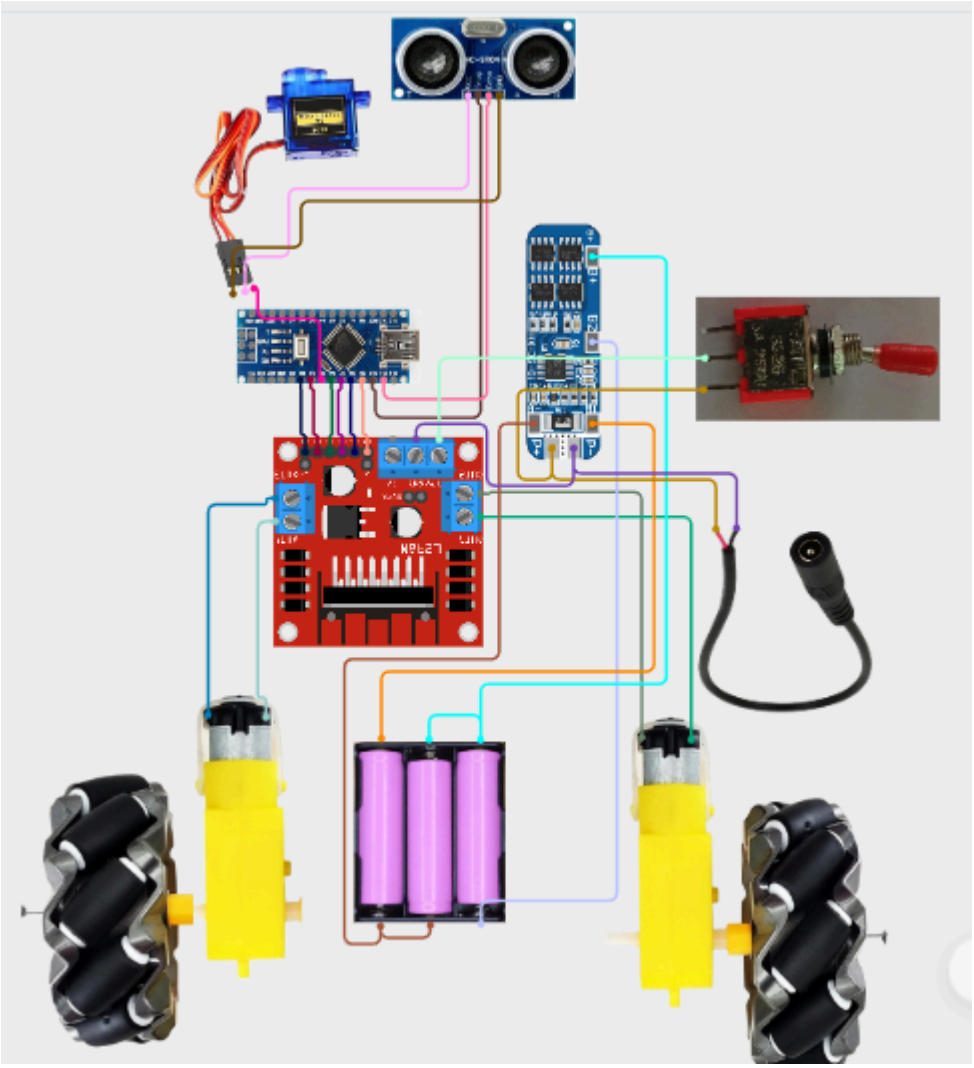
	IN4	**D8**	Right Motor Direction 2
HC-SR04 Ultrasonic	Trig	**D10**	Trigger Pulse Output
	Echo	**D11**	Echo Return Input
SG90 Servo Motor	Signal (Yellow)	**D3**	PWM Control Signal
	GND (Blue)	**GND**	Shared Common Ground

Source / Component	**Terminal / Wire**	**Connected To**	**Notes**
3S Li-ion Battery (with BMS)	Positive (+)	**L298N +12V Terminal**	Main battery supply voltage
	Ground (-)	**L298N GND Terminal**	Main power ground
L298N Motor Driver	GND Terminal	**Arduino UNO GND**	Establishes Common Ground
	+5V Out Terminal	**Arduino UNO VIN / 5V**	Regulated 5V to power Nano

	+5V Out Terminal	**SG90 VCC (Green)**	Regulated 4.9V–5V to power Servo
	+5V Out Terminal	**HC-SR04 VCC**	5V supply for Ultrasonic Sensor
	GND Terminal	**HC-SR04 GND**	Sensor Ground
DC Motors	OUT1 / OUT2	**Left DC Motor**	Motor A terminals
	OUT3 / OUT4	**Right DC Motor**	Motor B terminals

2. Power & Driver Pinout Connections3. Servo Wire Harness Color Pinout

Wire Color	**Function**	**Connection Point**
Yellow	Signal (PWM)	Arduino Nano **Pin D5**
Green	Power (+5V VCC)	L298N **5V Out Terminal**
Blue	Ground (GND)	Shared **Common Ground Rail**



Code overview:-

- **Ultrasonic Distance Measurement**: Emits a $10\mu\text{s}$ ultrasonic pulse from the HC-SR04 sensor and calculates distance based on the echo response time using the speed of sound. A $30ms$ timeout prevents code execution from freezing if no wave reflects back.
- **Continuous Monitoring**: The loop continuously measures distance straight ahead while driving forward, checking against a fixed safety threshold ($25cm$).
- **Obstacle Detection Sequence**: When an obstacle is detected within $25cm$, the robot immediately stops motor power, briefly reverses for $350ms$ to create maneuvering room, and comes to a complete halt.
- **Spatial Scanning**: The SG90 servo motor pans 20° to measure clearance on the right, resets to center, and then pans 160° to measure clearance on the left.

- **Directional Rerouting**: The microcontroller compares both distance readings (d_{right} vs. d_{left}). It executes a pivot turn toward the side with the larger open distance for $450ms$.
- **Differential Pivot Steering**: Turning is achieved by driving opposing wheels in reverse directions simultaneously, allowing zero-radius rotation without requiring forward driving space.

Future scope for upgrading and expanding autonomous obstacle-avoiding robot car:

Sensor Enhancement & Environment Mapping

- **LiDAR Integration**: Replacing or supplementing the HC-SR04 ultrasonic sensor with a compact 2D LiDAR module to create real-time 360-degree point-cloud maps of surrounding obstacles with millimeter accuracy.
- **IR Array / Cliff Sensors**: Adding bottom-facing infrared proximity sensors to detect edges, stairs, or table drop-offs to prevent the robot from falling.
- **Camera Module & Machine Vision**: Adding an ESP32-CAM or Raspberry Pi to enable computer vision for object classification, visual line-following, or traffic sign recognition using lightweight AI/ML models.

Navigation & Path Planning

- **SLAM (Simultaneous Localization and Mapping)**: Upgrading the main processor to run SLAM algorithms (such as Hector SLAM or Cartographer) so the robot can map unknown rooms and navigate efficiently from point A to point B without random scanning.
- **Wheel Encoders & Odometry**: Adding optical wheel encoders to measure motor rotations, enabling precise turning angles, dead reckoning, and closed-loop speed control.
- **IMU (Inertial Measurement Unit)**: Integrating an MPU-6050 (gyroscope + accelerometer) to correct drift during pivot turns and keep the robot moving in a straight line on uneven surfaces.

Control & Connectivity

- **Wireless Monitoring & Telemetry**: Integrating a Wi-Fi (ESP8266/ESP32) or Bluetooth (HC-05) module to send sensor telemetry, real-time battery voltage levels, and diagnostic data directly to a smartphone app or web dashboard.
- **Manual Override Mode**: Developing a dual-mode control system allowing seamless switching between fully autonomous obstacle avoidance and manual smartphone/joystick remote control.
- **GPS & Outdoor Navigation**: Adding a NEO-6M GPS module for outdoor waypoint navigation and geographic tracking.

Conclusion

This project successfully demonstrates the design, construction, and integration of an autonomous obstacle-avoiding robot car powered by an Arduino Nano and a 3S Li-ion battery system. By combining the real-time distance tracking of the HC-SR04 ultrasonic sensor with the panning capability of the SG90 servo motor, the vehicle effectively scans its surroundings to identify clear pathways and avoid collisions. The L298N motor driver enables precise differential steering and pivot maneuvers, allowing the robot to navigate around physical barriers without human intervention. Overall, the prototype serves as a functional foundation for mobile robotics, highlighting the practical application of embedded C programming, sensor integration, and automated path planning.